

Week 16

Cybersecurity Ethics

Ethical Issues in Cybersecurity

Ethical Framework in Cybersecurity

Ethical Best Practices in Cybersecurity

Dr. Ali Algarni

Progress

- **Quiz 4 (Socrative)???**
- **Tech Ignite 2025**
 - **Apr 29, 2025**
 - **Posters**
- **Final Exams**

Ethics (Recap)

- Morality and Ethics are used to distinguished between good and bad
- **Morality** related to personal and normative
- **Ethics** are Rules of conduct in a particular culture or group recognized by an external source or social system.
- **Ethical Theories** is the reference
- Ethics depends on place or domain (society and culture):
 - Theoretical level ('what is the best **theory** of the good life?')
 - Practical, applied level as will be our focus ('how **should we act in this** or that situation, based upon our best theories of ethics?').

The important ethical issues in cybersecurity

- **Morality** and **Ethics** are used to distinguished between good and bad
- **Morality** related to personal and normative
- **Ethics** are *Rules of conduct* in a particular culture or group recognized by an external source or social system.
- **Ethical Theories** is the reference and guidance
- Ethics depends on place or domain (society and culture):
 - Theoretical level ('what is the best **theory** of the good life?')
 - Practical, applied level as will be our focus ('how **should we act in this** or that situation, based upon our best theories of ethics?').

Ethics and cybersecurity

- **Ethical issues are at the core of cybersecurity practices**
- **A broader and better understanding of cybersecurity ethics** is essential to promoting and protecting human flourishing in an increasingly networked society.
- **No single set of ethical rules, guidelines, or insights can provide guidance in all cybersecurity circumstances**
 - adapted and applied to particular cybersecurity contexts and problems ‘in the wild.’
- **cybersecurity practices** are funding an increasing number of their own cybersecurity ethics-related programs and training tools
 - Appendix A

The important ethical issues in cybersecurity

- **Understanding life interests**
 - strong life interests in our health, happiness, family, friendship, social reputation, liberty, autonomy, knowledge, privacy, economic security, respectful and fair treatment by others, education, meaningful work, play, entertainment.
- Cybersecurity has a **broader ethical sweep**
 - It isn't just about technical issues like stopping hackers or fixing software bugs
 - Ethical concerns that affect society that impact privacy, freedom of speech, fairness, trust, safety, and even human rights.
 - not just about protecting data, but also about how, who, and what values we when protect data.

The important ethical issues in cybersecurity

Harms to Privacy

- Identity theft
- Blackmail and extortion
 - for example, using blackmail to pressure compromised employees to betray sensitive client information
- Corporate or government espionage and misconduct
 - messy, unorganized, and often overwhelming way data is created, shared, stored, and used all over the world.
 - ability to personally curate, delete, correct, or control data
 - thin, regionally inconsistent, and weakly enforced regulations reputational, economic, and emotional harms

The important ethical issues in cybersecurity

Harms to Property

- Property can be indirectly threatened by violations of data privacy, through
- Property can be directly targeted through cyber intrusions
 - May seek to misappropriate electronic funds, steal valuable intellectual property such as trade secrets, obtain bank account numbers
 - Remotely cause damage or destruction to an individual or organization's digital or physical property.
- Reasons include profit-seeking criminal enterprises
- Ethical harms of property cause harm for persons who rely
 - Secure good lives for themselves.
- cybersecurity professionals have a default ethical obligation
 - protect their organization's networks, clients, from any intrusions and attacks.

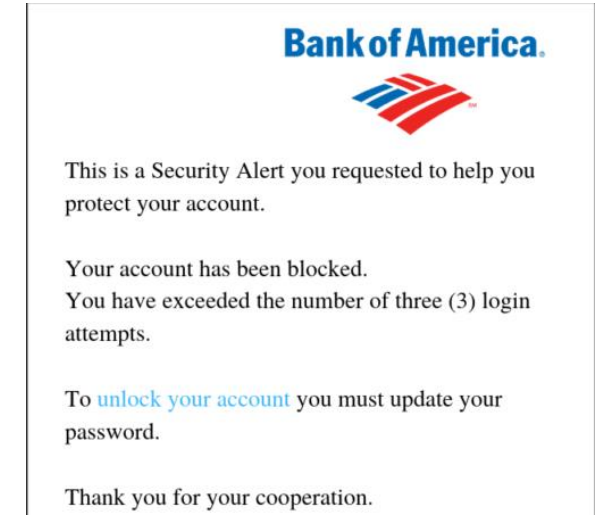
The Important Ethical Issues In Cybersecurity

Cybersecurity Resource Allocation

- The inevitably high cost of cybersecurity.
- Consume resources: time, money, and expertise
- Usability/product viability concerns can't justify weakening security.
 - Wi-Fi enabled pacemaker lack securing
- Not clear how the issue of resource allocation can be an ethical issue
 - right balance between security and other competing resource
 - Hospital attack: new and extremely time-consuming security logon vs many users and physician access
- Harms, benefits, rights and value should be balanced by cybersecurity staff before take decision.

The Important Ethical Issues In Cybersecurity

- **Transparency And Disclosure**
- Cybersecurity is a form of risk management
- Default ethical duty to disclose those risks when known, so that those affected can make informed decisions
- Significant debate over the proper method and scope of disclosure, as well as what constitutes a 'timely' notification.
 - Delay in notification until a patch is defensible.
 - Early notification may potentially invite an attack.
- Each cybersecurity scenario **involves different facts**,



The Important Ethical Issues In Cybersecurity

- **Cybersecurity Roles, Duties, And Interests**

The inevitably high cost of cybersecurity.

- Consume resources: time, money, and expertise
- Usability/product viability concerns can't justify weakening
- Not clear how the issue of resource allocation can be an ethical issue
 - right balance between security and other competing resource
 - Hospital attack: new and extremely time-consuming security logon vs many users and physician access
- Harms, benefits, rights and value should be balanced by cybersecurity staff before take decision

Common ethical challenges for cybersecurity professionals

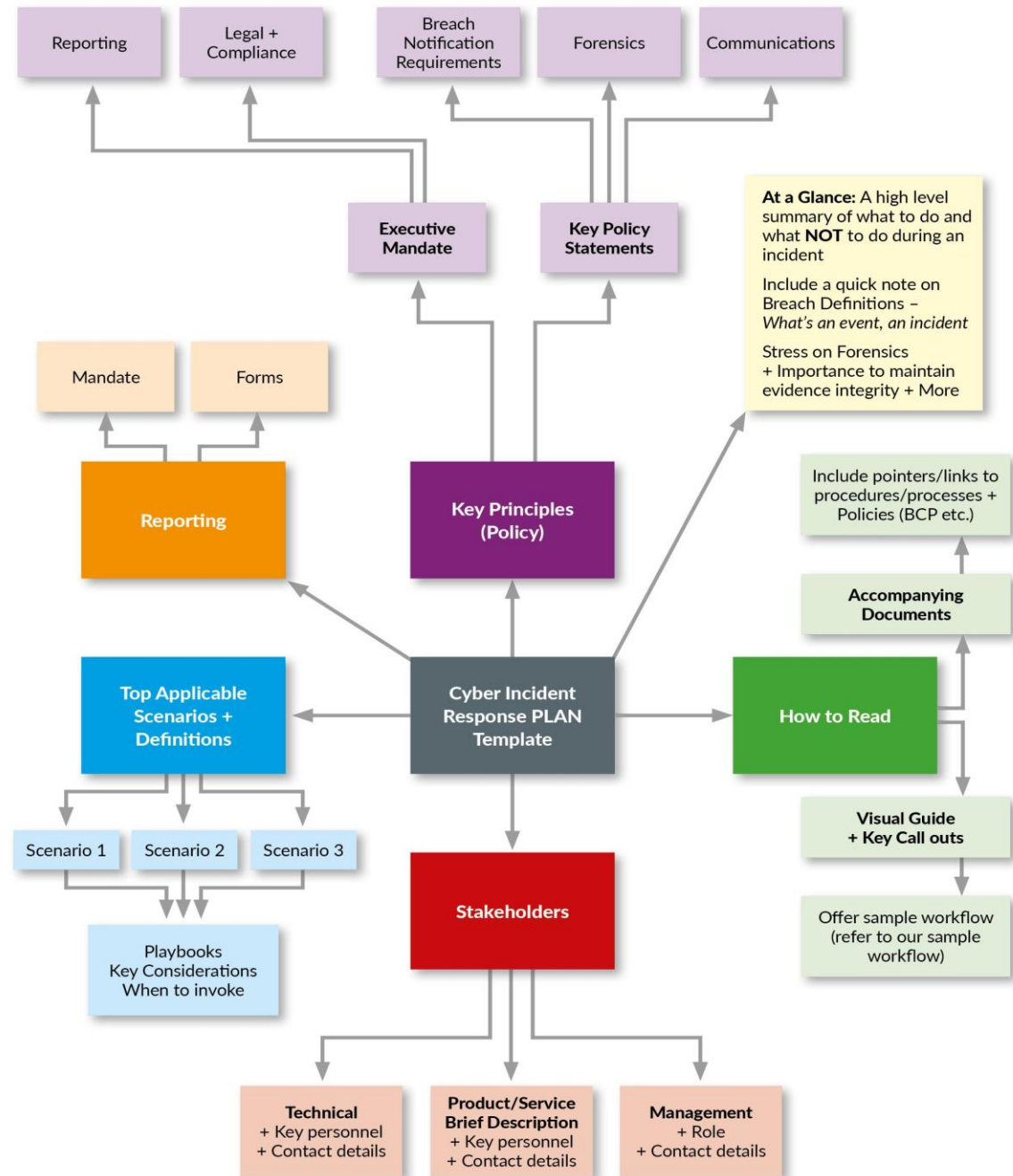
- Ethical Challenges In **Balancing Security With Other Values:**
- Ethical Challenges In **Threat/Incident Response:**
- Ethical Challenges In **Security Breach/Vulnerability:**
- Ethical Challenges In **Network Monitoring And User Privacy:**
- Ethical Challenges With **Competing Interests & Obligations:**
- Ethical Challenges In **Data Storage And Encryption:**
- Ethical Challenges In **IoT, Smart Grid, And Product Design:**
- Ethical Challenges With **Accountability For Cybersecurity:**
- Understanding **Broader Impacts Of Cybersecurity Practice**

Ethical Challenges In Balancing Security With Other Values

- Have we struck an ethically acceptable balance between cybersecurity and other values that may be in tension,
 - Network/device usability, reliability, speed, availability (CIA triad)
- Have we had honest conversations with our clients/customers/employers and users about how this balance will be struck?
 - A false sense of security to stakeholders? promises about security/system functionality but cannot keep?
- will we be able to justify our security resource allocations to affected stakeholders when there is serious breach of our system ?
- Do our actions reflect consistency, sincerity, and transparency in our value commitments.
 - upon inspection (like audit visit), will they reveal arbitrary, inconsistent, insincere, or deceptive professions of value?

Ethical Challenges In Threat/Incident Response

- Appropriate incident response plan, for each type of threat or incident that we anticipate facing?
 - concrete action plans for the worst-case-scenarios (e.g. mitigation strategies to limit or remedy harms to others if breach/attack fall short)
- Adequate resources and systems in place to successfully implement our incident response plan?
 - plan only reflect what we'd like to be able to do in response to a threat or security breach, not what we actually can and are prepared to do? gap ethically acceptable?
- Ethical gray areas we face in our incident response practice?
 - How aggressive can we justifiably be in repelling and discouraging attacks
- How will we respond to ransomware attacks affecting our network or users?
 - ethically right to pay ransoms to attackers to restore system functionality or access



Ethical Challenges In Security Breach/Vulnerability:

- **Ethical plan for when and how to notify network/software users and other stakeholders of security incidents, including breaches and vulnerabilities?**
 - What ethical conditions must be met in order to justify a delay of notification?
- **How do we meet the need for reporting that is accurate, timely, and helpful?**
 - Do we have an efficient system in place for rapidly investigating and planning an effective response to an incident?
- **Have we considered not only our own perspective, but the likely perspective of other stakeholders?**
 - Is a delay or selective notification of a breach or vulnerability likely to cause damage downstream to public, client, or user trust in our services, our product, or our organizational values even the plan is ethically sound?
- **Have we identified the ethically appropriate balance between overreaction to breaches and vulnerabilities and underreaction?**
 - Is our team predisposed to dismiss signs of a possible breach as false positives, or to discount the true risk to others posed by a vulnerability?

Ethical Challenges in Network Monitoring and User Privacy.

- How to monitor our network effectively without making unjustifiable intrusions upon users and their privacy?
- How should we monitor and control network activity on personal devices such as laptops and cell phones not owned by our organization and legitimately used for many other purposes.
- To what extent should users of the network be made aware of our security monitoring activities?

Ethical Challenges In Data Storage And Encryption

- How can we responsibly and safely store and transmit sensitive information?
 - Clients/ customers given clear and accurate information about our encryption
- Do we take adequate care in contracting with third parties
- What are the ethical risks and benefits associated with various forms and strengths of encryption we might use?
- How will we respond to requests from law-enforcement or intelligence agencies to weaken our encryption practices or decrypt specific devices?
- What are the ethical risks of long-term data storage? How long we are justified in keeping sensitive data, and when/how often should it be purged for security purposes?

Ethical Challenges With Accountability For Cybersecurity

- Who is designated as responsible for each aspect of cybersecurity practice? avoid a scenario of ethical cybersecurity practice is a high-level goal, and no specific individuals are tasked?
 - Assign Responsibility Explicitly, Incorporate Ethics into Processes, training tools,
- Who should and will be held accountable for various risks or harms that might be imposed on others by our cybersecurity practice?
 - problem of many hands
- What organizational/team policies and routines must we establish in advance and enforce ethical cybersecurity practice?
 - For example, regularly scheduled security audits, incident reports, reviews of access policies, privilege levels, password management and supervision routines
- Have we placed cybersecurity efforts in appropriately skilled and responsible hands, with appropriate levels of instruction, training, and guidance?
 - Offload too much responsibility for sound security practice

Ethical Challenges With Accountability For Cybersecurity

- Who is designated as responsible for each aspect of cybersecurity practice? avoid a scenario of ethical cybersecurity practice is a high-level goal, and no specific individuals are tasked?
 - Assign Responsibility Explicitly, Incorporate Ethics into Processes, training tools,
- Who should and will be held accountable for various risks or harms that might be imposed on others by our cybersecurity practice?
 - problem of many hands
- What organizational/team policies and routines must we establish in advance and enforce ethical cybersecurity practice?
 - For example, regularly scheduled security audits, incident reports, reviews of access policies, privilege levels, password management and supervision routines
- Have we placed cybersecurity efforts in appropriately skilled and responsible hands, with appropriate levels of instruction, training, and guidance?
 - Offload too much responsibility for sound security practice

Why Do Cybersecurity Professionals Have Obligations To The Public?

- Co-exist with ethical obligations to one's family, friends, employer, local community, and even oneself.
- Some cybersecurity contexts do involve life and death risks
- Risks of harm to the public that we must guard against
- It remains true that the nature and details of our obligations to the public as cybersecurity professionals can be unclear
- Stakeholder impact on the firm/institution

Cybersecurity Professionals And The Public Good

- Professional is not just job
 - express a belief, conviction, value or promise to a general audience that you expect that audience to hold you accountable
- **Public good** refers to actions, services, or outcomes that benefit society as a whole, rather than just individual organizations or clients.
- Professionals secure a vital public good.
- Ensure data privacy and protection
- Report and mitigate threats
- Educate and raise awareness

Distinguishing And Ranking Competing Stakeholder Interests

- Variety of stakeholders potentially impacted by my action, whose interests are not always aligned.
- Distinguishing them by the different 'stakes' they hold in what we do as cybersecurity practitioners
- Ethical decisions isn't just about following rules or personal preferences, it's about thoughtfully considering how your actions affect all the people who may be impacted

Best Practices For Cybersecurity Ethics

- Keep Cybersecurity Ethics in the Spotlight—and Out of the Compliance Box
- Consider the Human Lives and Interests Behind systems.
- Establish Chains of Ethical Responsibility and Accountability
- Focus on Downstream Risks and Uses of Cybersecurity practice.
- Practice Cybersecurity Disaster Planning and Crisis Response:
- Promote Values of Transparency, Autonomy, and Trustworthiness: .
- Establish Chains of Ethical Responsibility and Accountability.
- Design for Privacy and Security
- Invite Diverse Stakeholder Input:

Best Practices For Living Well

- Look for Moral Exemplars
- Exercise Moral Imagination:
- Practice Self- Reflection/Examination
- Acknowledge Our Own Moral Strength
- Seek the Company of Other Moral Persons